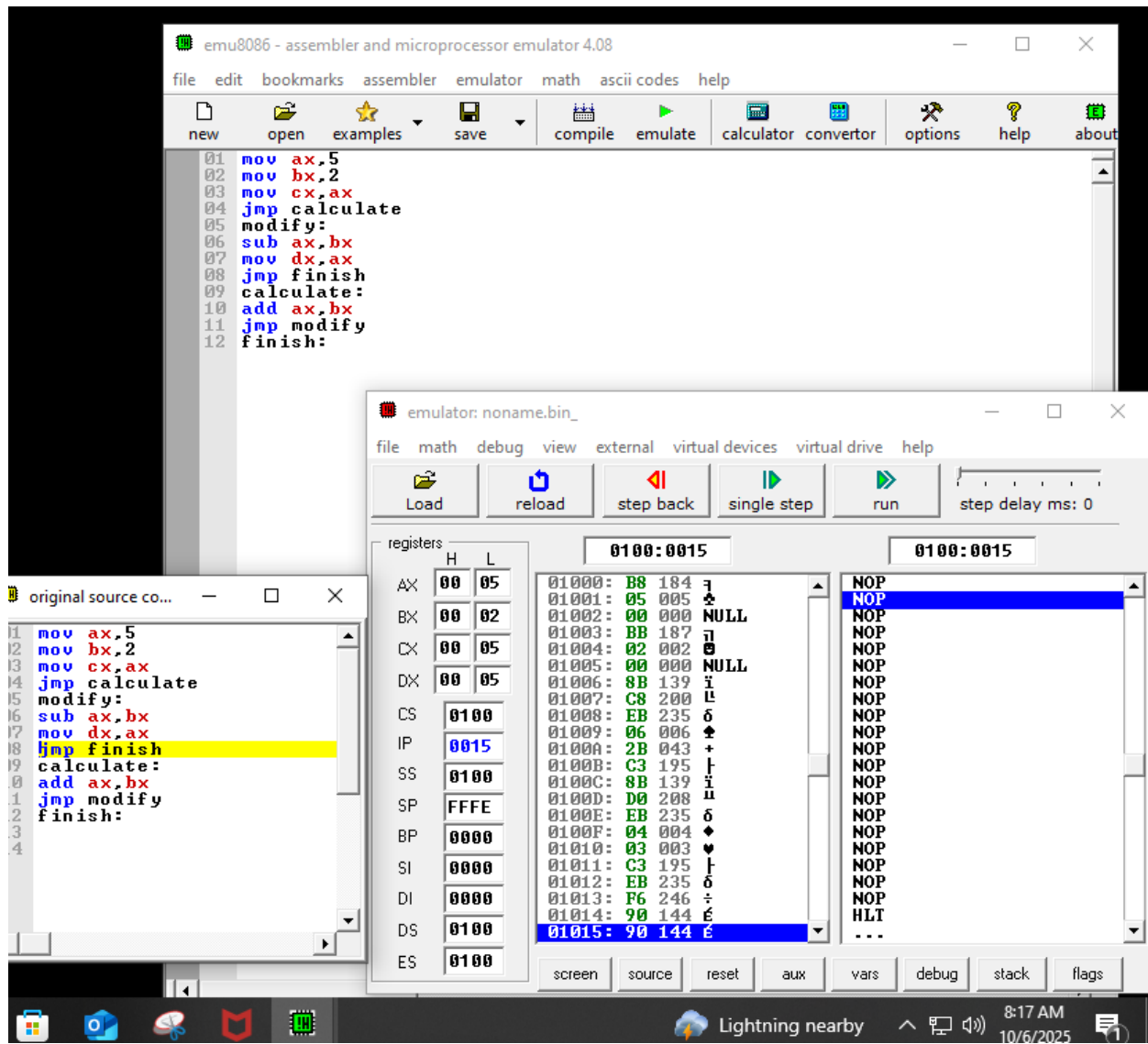


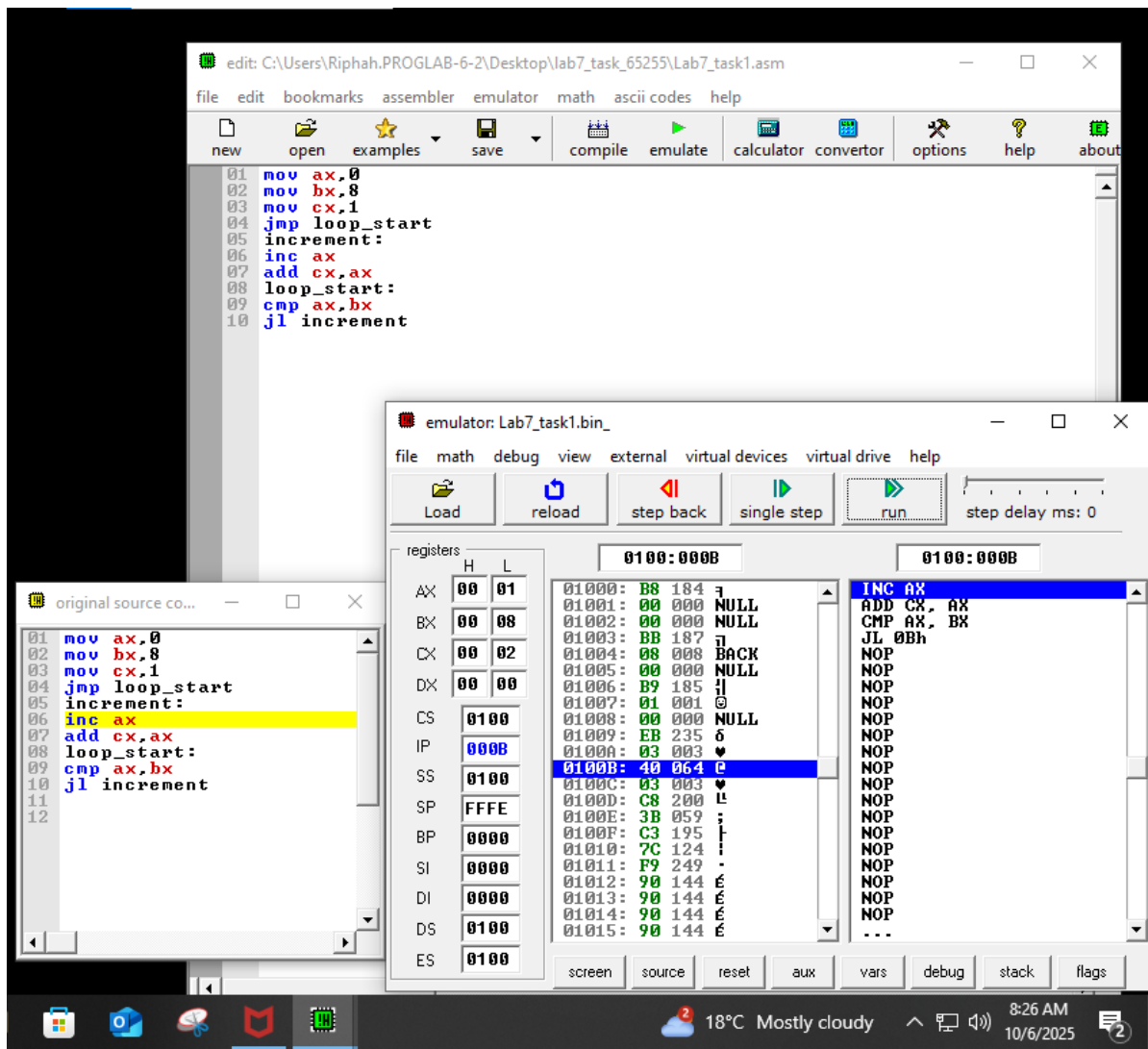
COAL

LAB 7 TASKS

TASK 1



Task 2



Task 3

The image shows a screenshot of an x86 assembly editor and emulator interface. The main window is titled "edit: C:\Users\Riphah.PROGLAB-6-2\Desktop\lab7_task_65255\Lab7_task2.asm". It contains assembly code for a division routine. The code is as follows:

```
01 mov ax,10
02 mov bx,3
03 mov dx,0
04 jmp divison
05 done:
06 mov cx,ax
07 mov ax,dx
08 jmp end
09 divison:
10 sub ax,bx
11 inc dx
12 cmp ax,bx
13 jge divison
14 jmp done
15 end:
```

On the left, there is a "flags" panel with the following values:

- CF: 1
- ZF: 0
- SF: 1
- OF: 0
- PF: 0
- AF: 1
- IF: 1
- DF: 0

Below the flags panel is an "analyse" button. At the bottom left, there is a window titled "original source co..." showing the same assembly code. The "original source co..." window has the following code:

```
01 mov ax,10
02 mov bx,3
03 mov dx,0
04 jmp divison
05 done:
06 mov cx,ax
07 mov ax,dx
08 jmp end
09 divison:
10 sub ax,bx
11 inc dx
12 cmp ax,bx
13 jge divison
14 jmp done
15 end:
16
17
```

The main emulator window is titled "emulator: Lab7_task2.bin_". It has a menu bar with "file", "math", "debug", "view", "external", "virtual devices", "virtual drive", and "help". The toolbar includes buttons for "Load", "reload", "step back", "single step", "run", and a "step delay ms: 0" slider. The "registers" panel shows the following values:

Register	H	L
AX	00	03
BX	00	03
CX	00	01
DX	00	03
CS	0100	
IP	001A	
SS	0100	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0100	
ES	0100	

The memory dump shows the following values:

Address	Value	Comment
0100B: 8B	139	i
0100C: C8	200	U
0100D: 8B	139	i
0100E: C2	194	T
0100F: EB	235	δ
01010: 09	009	TAB
01011: 2B	043	+
01012: C3	195	↑
01013: 42	066	B
01014: 3B	059	;
01015: C3	195	↑
01016: 7D	125	>
01017: F9	249	-
01018: EB	235	δ
01019: F1	241	±
0101A: 90	144	E
0101B: 90	144	E
0101C: 90	144	E
0101D: 90	144	E
0101E: 90	144	E
0101F: 90	144	E
01020: 90	144	E

The memory dump also shows a list of instructions: NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, NOP, HLT, ...

The taskbar at the bottom shows the system clock as 8:45 AM on 10/6/2025, along with other icons like USD/JPY +1.72% and a notification icon.

Task 4

edit: C:\Users\Riphah.PROGLAB-6-2\Desktop\lab7_task_65255\Lab7_task3.asm

file edit bookmarks assembler emulator math ascii codes help

new open examples save compile emulate calculator convertor options help about

01 mov ax,9
02 mov bx,ax
03 dec bx
04 jmp check
05 process:
06 dec ax
07 add cx,ax
08 check:
09 cmp ax,0
10 jg process
11 mov dx,cx

flags

CF 0
ZF 1
SF 0
OF 0
PF 1
AF 0
IF 1
DF 0

analyse

original source co...

01 mov ax,9
02 mov bx,ax
03 dec bx
04 jmp check
05 process:
06 dec ax
07 add cx,ax
08 check:
09 cmp ax,0
10 jg process
11 mov dx,cx
12
13

emulator: Lab7_task3.bin_

file math debug view external virtual devices virtual drive help

Load reload step back single step run step delay ms: 0

registers

	H	L
AX	00	00
BX	00	08
CX	00	24
DX	00	24
CS	0100	
IP	0026	
SS	0100	
SP	FFFE	
BP	0000	
SI	0000	
DI	0000	
DS	0100	
ES	0100	

0100:0026

01016: 90 144 €
01017: 90 144 €
01018: 90 144 €
01019: 90 144 €
0101A: 90 144 €
0101B: 90 144 €
0101C: 90 144 €
0101D: 90 144 €
0101E: 90 144 €
0101F: 90 144 €
01020: 90 144 €
01021: 90 144 €
01022: 90 144 €
01023: 90 144 €
01024: 90 144 €
01025: 90 144 €
01026: F4 244 ↑
01027: 00 000 NULL
01028: 00 000 NULL
01029: 00 000 NULL
0102A: 00 000 NULL
0102B: 00 000 NULL

0100:0026

NOP
NOP
NOP
NOP
HLT
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
ADD [BX + SI], AL
...

screen source reset aux vars debug stack flags

GBP/USD -0.24% 9:29 AM 10/6/2025

Task 5

The screenshot displays an x86 assembly editor and emulator interface. The main window shows the assembly code for a Fibonacci sequence calculator. The code is as follows:

```
01 mov ax,0
02 mov bx,1
03 mov cx,7
04 jmp fibonacci
05 stop:
06 mov dx,bx
07 hlt
08 fibonacci:
09 mov si,ax
10 mov ax,bx
11 add bx,si
12 dec cx
13 jnz fibonacci
14 jmp stop
```

On the left, a 'flags' panel shows the status of various flags: CF, ZF, SF, OF, PF, AF, IF, and DF. The IF flag is set to 1. Below the flags panel is an 'analyse' button.

The bottom-left window shows the original source code with line numbers 01 through 16. Line 12, 'dec cx', is highlighted in yellow.

The bottom-right window is the emulator, titled 'emulator: Lab7_task4.bin_'. It features a menu bar (file, math, debug, view, external, virtual devices, virtual drive, help) and a toolbar with buttons for Load, reload, step back, single step, run, and a step delay slider. The registers panel shows the current state of the CPU registers:

Register	H	L
AX	00	08
BX	00	0D
CX	00	02
DX	00	00
CS	0100	
IP	0014	
SS	0100	
SP	FFFE	
BP	0000	
SI	0005	
DI	0000	
DS	0100	
ES	0100	

The memory window shows the current instruction being executed at address 0100:0014:

Address	Hex	Dec	Comment
01000:	B8	184	MOV SI, AX
01001:	00	000	MOV AX, BX
01002:	00	000	ADD BX, SI
01003:	BB	187	DEC CX
01004:	01	001	JNE 0Eh
01005:	00	000	JMP 010Bh
01006:	B9	185	NOP
01007:	07	007	NOP
01008:	00	000	NOP
01009:	EB	235	NOP
0100A:	03	003	NOP
0100B:	8B	139	NOP
0100C:	D3	211	NOP
0100D:	F4	244	NOP
0100E:	8B	139	NOP
0100F:	F0	240	NOP
01010:	8B	139	NOP
01011:	C3	195	NOP
01012:	03	003	NOP
01013:	DE	222	NOP
01014:	49	073	NOP
01015:	75	117	NOP

The taskbar at the bottom shows the system clock as 9:46 AM on 10/6/2025, along with other system icons and a GBP/USD exchange rate of -0.23%.